Deformation Theory

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1 Introduction

In the section, we will introduce the definitions and intuition for first order deformations.

1.1 Intuition

Embedded Deformation: Suppose we have a smooth submanifold X embedded an ambient complex manifold Y. The embedding is equipped with a normal bundle $N_Y X$. By the tubular neighborhood theorem, we have a embedding of its total space:

$$\begin{tikzcd} X \arrow[rr, hook] \arrow[dr, "\text{0-section}"'] & & Y \\ & N_Y X \arrow[ur, hook, dotted] & \end{tikzcd}$$

A smooth deformation of Y inside X is then a smooth section of $N_Y X$: at each point $x \in X$, the section give you the normal direction along which to "infinitesimally" deform X inside Y. This definition offers some differential topologically intuition, even though we no longer have an analog of the tubular neighborhood theorem in the holomorphic/algebraic setting.

More generally, we are given the data of

1. A morphism of objects $f : X \to Y$ in some category (e.g. schemes, complex manifolds, etc).
2. An "infinitesimal thickening" of X and Y, which are prescribed injective morphisms $X \to X'$ and $Y \to Y'$.

A deformation of f is then a lift of the morphism $X \to Y$ to a morphism $X' \to Y'$

$$\begin{tikzcd} X \arrow[r] \arrow[d] & Y \arrow[d] \\ X' \arrow[r, dotted, "?"'] & Y' \end{tikzcd}$$

Deformation of Complex Structure: Viewing a complex manifold X as a real manifold with an integrable almost complex structure J, a deformation of the complex structure is a family of almost complex structures J_t parametrized by $t \in (-\epsilon, \epsilon)$ such that $J_0 = J$ and each J_t is integrable. Infinitesimally, we can think of this as a perturbation of the almost complex structure J by a small parameter t in the space of endomorphisms of the tangent bundle TX that satisfy $J_t^2 = -\mathrm{Id}$ and the integrability condition (vanishing of the Nijenhuis tensor).

The theory of Kodaira-Spencer describes how such deformations can be understood in terms of certain cohomology groups associated with the manifold. More precisely, first order deformations of the complex structure on X are classified by the cohomology group $H^1(X, T_X)$, where T_X is the holomorphic tangent bundle of X.

Deformation and Moduli Problem In relation to deformation of complex structure, suppose we have concretely constructed the moduli space $\mathcal{M}_g$ parameterizing smooth projective curves of genus g. A point $[C] \in \mathcal{M}_g$ corresponds to an isomorphism class of a smooth projective curve C of genus g. A deformation of the curve C can be thought of as a small perturbation of the complex structure on C, leading to a family of curves C_t parameterized by a small parameter t. Infinitesimally, this corresponds to moving along a tangent vector at the point $[C]$ in the moduli space $\mathcal{M}_g$. The tangent space at $[C]$ can be identified with the first cohomology group $H^1(C, T_C)$, where T_C is the holomorphic tangent bundle of the curve C.

Suppose we have a fine moduli space $\mathcal{M}$ parameterizing certain objects, meaning any family of such objects over a base B is given by a morphism

$$B \to \mathcal{M}$$

Then an infinitesimal deformation of an object X corresponding to a point $[X] \in \mathcal{M}$ is given by a family over the dual numbers $\mathrm{Spec}(\mathbb{C}[\epsilon]/(\epsilon^2))$ (we can think of the dual numbers as a point equipped with a tangent vector), such that the fiber over 0 is isomorphic to X. Then, the classifying map

$$T_0 B \to T_{[X]}\mathcal{M}$$

precisely specifies a tangent vector at $[X]$.

2 First Order Deformation

Definition 2.0.1. The **dual numbers** over a field k is the ring

$$D := k[t]/(t^2)$$

Note that $\mathrm{Spec}(D)$ is the one-point space consisting of the prime ideal (t). Its Zariski tangent space is given by

$$T_{(t)}D := \mathrm{Hom}_k((t)/(t^2), k)$$

which is a one-dimensional k-vector space. In contrast, $\mathrm{Spec}(k)$ is also a one-point space, but its Zariski tangent space is zero-dimensional. Thus, we can think of $\mathrm{Spec}(D)$ as a point equipped with a tangent vector. The dual numbers will be our base space for first order deformations.

Definition 2.0.2. Let Y be a scheme over k, and X be a closed subscheme. A **first order embedded deformation** of X in Y is a closed subscheme $X' \subset Y \times_k \mathrm{Spec}(D)$ such that

1. X' is flat over D
2. The fiber over the point is X, i.e
$$X' \times_{\mathrm{Spec}(D)} \mathrm{Spec}(k) = X$$

We would like to classify all such deformations. It is useful to characterize flatness over the dual numbers first:

Lemma 2.1. A module M over a commutative ring R is flat iff for every ideal $I \subset R$, the module homomorphism

$$I \otimes_R M \to M$$

is injective.

Proof. Follows directly from the fact that flatness is equivalent to the vanishing of $\mathrm{Tor}_1^R(R/I, M)$ for all ideals $I \subset R$. ☐

2.1 Embedded Deformation of Schemes

We first deal with the affine case. Suppose $X = \mathrm{Spec}(B)$ is affine, and Y is a closed subscheme defined by an ideal $I \subset B$. We see that first-order embedded deformations of Y in X correspond to ideals $J \subset B' := B[t]/(t^2)$ such that

1. B'/J is flat over D
2. The image of J under the quotient map
$$B[t]/(t^2) \to B$$
is I.

Proposition 2.1.1. There is a bijection between the set of first-order embedded deformations of Y in X and $\mathrm{Hom}_B(I, B/I)$.

Proof. Suppose we are given an ideal $J \subset B'$ such that the J mod $t = I$. By Lemma 2.1, flatness of B'/J over D is equivalent to the exactness of

$$0 \to B'/J \otimes_D (t) \to B'/J \to B'/J \otimes_D k \to 0$$

Viewing (t) as a k-module, we have

$$B/I \otimes_k (t) \cong (B'/J \otimes_D k) \otimes_k (t) \cong B'/J \otimes_D (t)$$

So we may rewrite the exact sequence as

$$0 \to B/I \xrightarrow{\cdot t} B'/J \to B/I \to 0$$

Since $B' := B[t]/(t^2)$ splits as $B \oplus tB$ as B-modules, for every $f \in I$, there exist lifts of the form $f + tg \in J$, where $f, g \in B$. By exactness of the sequence above, different lifts differ by an element in tI, so we may then define a function

$$\varphi : I \to B/I$$

by $\varphi(f) = \overline{g}$. It is easy to check this is well-defined and a B-module homomorphism. Conversely, given a B-module homomorphism $\phi : I \to B/I$, we may define an ideal:

$$J := \{f + tg \mid f \in I, g \in B, g \bmod I = \phi(f)\}$$

Flatness of B'/J over D can be checked by by the same exact sequence as above: an element tb is in the ideal J iff $b \in I$. The two constructions are inverse to each other, and we have the desired bijection.

□

Definition 2.1.1. The first order embedded deformations of Y in X is **trivial** if they correspond to the zero homomorphism in $\mathrm{Hom}_B(I, B/I)$, which corresponds to the ideal $J = I \oplus tI \subset B[t]/(t^2)$.

Example 2.1.1. We are given curve C carved out by $f(x, y)$, embedded in $\mathbb{A}^2$ via the canonical map

$$k[x, y] \to k[x, y]/(f)$$

By our above analysis, first order embedded deformations of C in $\mathbb{A}^2$ correspond to elements of

$$\mathrm{Hom}_{k[x,y]/(f)}((f), k[x, y]/(f)) \cong k[x, y]/(f)$$

In particular, each $g \in k[x, y]/(f)$ gives rise to a first order deformation defined by the principal ideal $J = (f + tg)$. We can think of this as perturbing f by g infinitesimally.

Recall that for a closed embedding of schemes $X \to Y$ defined by an ideal sheaf $\mathcal{I} \subset \mathcal{O}_Y$, the normal sheaf $\mathcal{N}_{X/Y}$ is defined as the dual of the conormal sheaf $\mathcal{I}/\mathcal{I}^2$. Moreover,

$$\mathrm{Hom}_Y(\mathcal{I}, \mathcal{O}_X) \cong \mathrm{Hom}_X(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_X)$$

Since flatness is a local property, and our construction above is natural and compatible with localization, we may globalize Proposition 2.1.1 to obtain:

Theorem 2.2. Let Y be a scheme over a field k, and let X be a closed subscheme of Y. Then the deformations of X over D in Y are in natural one-to-one correspondence with elements of $H^0(X, N_{X/Y})$, with the zero element corresponding to the trivial deformation.

2.2 Deformation of Coherent Sheaves

Let X be a scheme over a field k, and $\mathcal{F}$ be a coherent sheaf on X. A first order deformation of $\mathcal{F}$ is a coherent sheaf $\mathcal{F}'$ on $X' := X \times_k \mathrm{Spec}(D)$ that is

1. flat over D;

2. equipped with a homomorphism $\mathcal{F}' \to \mathcal{F}$ such that the induced map $\mathcal{F}' \otimes_D k \to \mathcal{F}$ is an isomorphism.

Two such deformations $\mathcal{F}', \mathcal{F}''$ are considered isomorphic if there exists an isomorphism $\mathcal{F}' \to \mathcal{F}''$ compatible with their morphisms to $\mathcal{F}$.

Theorem 2.3. Let X be a scheme over a field k, and $\mathcal{F}$ be a coherent sheaf on X. Then the isomorphism classes of first order deformations of $\mathcal{F}$ are in bijection with elements of $\mathrm{Ext}^1_X(\mathcal{F}, \mathcal{F})$, with the zero element corresponding to the trivial deformation.

Proof. By Lemma 2.1, a first order deformation $\mathcal{F}'$ of $\mathcal{F}$ fits into a short exact sequence of D-modules

$$0 \to \mathcal{F} \xrightarrow{t} \mathcal{F}' \to \mathcal{F} \to 0$$

Note that tensoring the split exact sequence

$$0 \to k \to D \to k \to 0$$

with $\mathcal{O}_X$, we get canonical splitting $\mathcal{O}_X \to \mathcal{O}'_X$, so we may view $\mathcal{F}'$ as an $\mathcal{O}_X$-module. Thus, the above short exact sequence of $\mathcal{O}_X$-modules is an element of $\mathrm{Ext}^1_X(\mathcal{F}, \mathcal{F})$. Conversely, given an extension, we may define a $\mathcal{O}_{X'}$-module structure on it by letting t act via first projecting along $\mathcal{F}' \to \mathcal{F}$, and then apply $\mathcal{F} \xrightarrow{t} \mathcal{F}'$. The two constructions are inverse to each other, giving the desired bijection.

□

We may apply the theorem to the special case of vector bundles, i.e locally free sheaves.

Corollary 2.3.1. The first order deformation of a vector bundle $\mathcal{E}$ on X are classified by elements of $H^1(X, \mathcal{E}nd(\mathcal{E}))$, with the trivial deformation corresponding to the zero element.

Proof. Note that $H^1(X, \mathcal{E}nd(\mathcal{E})) \cong \mathrm{Ext}^1(\mathcal{O}_X, \mathcal{E}nd(\mathcal{E}))$, since $\Gamma(X, -) = \mathrm{Hom}(\mathcal{O}_X, -)$ so their derived functors are isomorphic. By [2], Prop 6.7, we have the adjunction

$$\mathrm{Ext}^1(\mathcal{O}_X, \mathcal{E}nd(\mathcal{E})) \cong \mathrm{Ext}^1(\mathcal{E}, \mathcal{E})$$

. □

Example 2.3.1. We know that the moduli functor of vector bundles on a scheme X is not representable, i.e there is no fine moduli space of vector bundles over X. One may see this via the existence of non-trivial automorphisms of trivial vector bundle, as we can glue trivial families together and get a non-trivial family with isomorphic fibers. Yet, we can still study the local structure of the moduli problem via deformation theory: Let $X = \mathbb{P}^1$, and $\mathcal{E} = \mathcal{O}(1) \oplus \mathcal{O}(-1)$. It is easy to compute that $\mathcal{E}nd(\mathcal{E}) \cong \mathcal{O}(2) \oplus \mathcal{O}(-2) \oplus 2$. By Corollary 2.3.1, we see the deformations of $\mathcal{E}$ are in bijection with elements of

$$H^1(\mathbb{P}^1, \mathcal{E}nd(\mathcal{E})) \cong k$$

coming from the $\mathcal{O}(-2)$ summand. The non-trivial infinitesimal deformation can be seen in the family given by extensions

$$0 \to \mathcal{O}(-1) \to \mathcal{E}_t \to \mathcal{O}(1) \to 0$$

parametrized by $t \in \mathrm{Ext}^1(\mathcal{O}(-1), \mathcal{O}(1)) \cong k$. By Grothendieck's theorem, any vector bundle on $\mathbb{P}^1$ splits as a direct sum of line bundles, and the additivity of Chern class forces $\mathcal{E}_t \cong \mathcal{O}(-d) \oplus \mathcal{O}(d)$. Moreover, $\mathcal{O}(-1)$ admits a non-zero morphism to $\mathcal{O}(k)$ iff $k \geq -1$, so the only possibilities are $d = 0, 1$. The case where $d = 1$ corresponds to the trivial extension $\mathcal{E}_0 = \mathcal{E}$, and the case $d = 0$ gives us the non-trivial extension when $t \neq 0$.

3 Cohomology of Commutative Rings

We are now interested in studying more general deformations of schemes, not necessarily embedded in an ambient space.

Definition 3.0.1. Let X be a k-scheme, and A an Artin ring over k. A **deformation** of X over A is a scheme X' flat over A, together with a closed immersion $i : X \to X'$ such that the induced map

$$i \times_A k : X \to X' \times_A k$$

is an isomorphism. Tow such deformations are **equivalent** if there is an isomorphism $\phi : X' \to X''$ compatible with the closed immersions of X.

We then recall some facts on Kähler differentials. Given a morphism of commutative rings $A \to B$, the relative Kähler differentials $\Omega_{B/A}$ is defined to be the B-module generated by symbols db for $b \in B$, subject to the relations

- $da = 0$,
- $d(a_1 + a_2) = da_1 + da_2$,
- $d(ab) = a(db) + b(da)$

The construction is compatible with localization, so we may globalize to general schemes. Given a morphism of schemes

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

there is a natural exact sequence of sheaves on X:

$$f^*\Omega_{Y/Z} \to \Omega_{X/Z} \to \Omega_{X/Y} \to 0$$

and we wish to extend such exact sequences to the left. Historically, Lichtenbaum and Schleissinger directly constructed the first 3 functors T^0, T^1, T^2 that continued the long exact sequence; later André and Quillen generalized the construction to all T^i using simplicial methods. In particular, Quillen in [4] was able to develop the theory in the general framework of model categories in which derived functors can be defined, even when the underlying category is no longer abelian.

3.1 The Cotangent Complex

Definition 3.0.2. In a category $\mathcal{C}$, an object c in called **abelian** if $\mathrm{Hom}_{\mathcal{C}}(-, c)$ carries the structure of an abelian group.

Remark 3.0.1. For a commutative ring A, the category of commutative A-algebras $\mathbf{Alg}_A$ does not have any abelian objects: sum of two unital ring homomorphisms is never unital. Instead, we have to consider the relative situation.

For the rest of the section, let B be a fixed A-algebra, and $\mathcal{C} := (\mathbf{Alg}_A)/B$ be the corresponding over category. Consider the functor

$$B \ltimes - : \mathbf{Mod}_B \to \mathcal{C}$$

where for a B-module M, the module $B \oplus M$ is given the A-algebra structure

$$(a, x)(b, y) = (ab, bx + ay)$$

with A-acting diagonally. It is canonically equipped with a A-algebra projection map to B, so we have a well-defined object in $\mathcal{C}$, denoted by $B \ltimes M$. It is straightforward to check the construction is functorial.

Lemma 3.1. There is a natural isomorphism

$$\mathcal{C}(X, B \ltimes M) \cong \mathrm{Der}_A(X, M)$$

Proof. By construction, a morphism $f \in \mathcal{C}(X, B \ltimes M)$ is a diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\epsilon"'] & B \ltimes M \arrow[dl, "\pi"] \\ B & \end{tikzcd}$$

where ϵ is the structure augmentation map, and π is the canonical projection. We see that f must factor as $\epsilon \oplus f'$ for some A-module homomorphism $f' : X \to M$. Viewing M as an X-module via ϵ, we see f' also satisfies the Leibniz rule:

$$f'(xy) = \epsilon(x)f'(y) + \epsilon(y)f'(x)$$

Thus, $f' \in \mathrm{Der}_A(X, M)$. Conversely, given a derivation $d \in \mathrm{Der}_A(X, M)$, the A-module homomorphism $\epsilon \oplus d : X \to B \oplus M$ is also a ring homomorphism, and defines an element in $\mathcal{C}(X, B \ltimes M)$. The two constructions are inverse to each other, giving the desired isomorphism. $\square$

The lemma implies that the image of the functor $B \ltimes -$ lands in the $\mathcal{C}_{ab}$, the subcategory of abelian objects in $\mathcal{C}$.

Proposition 3.1.1. The functor

$$B \ltimes - : \mathbf{Mod}_B \to \mathcal{C}_{ab}$$

is an equivalence of categories. In particular, $\mathcal{C}_{ab}$ is an abelian category.

Proof. We only need to construct the inverse: an object X in $\mathcal{C}_{ab}$ is automatically a group object in $\mathcal{C}$, so that we have an unit $e : B \to X$ and a multiplication map $m : X \times_B X \to X$ satisfying the group axioms. In particular, the unit give us a splitting

$$B \xrightarrow{e} X \xrightarrow{\epsilon} B$$

so that $X \cong B \oplus M$ as B-modules for the B-module $M = \ker(\epsilon)$. The identity axiom gives us the following diagram:

$$\begin{tikzcd} X \arrow[r, "(Id, e\epsilon)"] \arrow[d, "(e\epsilon, Id)"'] \arrow[dr, "Id"] & X \times_B X \arrow[d, "m"] \\ X \times_B X \arrow[r, "m"'] & X \end{tikzcd}$$

Note that $e\epsilon$ is identically 0 on $M \subset X$, and therefore $m(x, 0) = x$ and $m(0, y) = y$ for all $x, y \in M$. Finally, the fact that m is a ring homomorphism gives

$$xy = m(x, 0)m(0, y) = m((x, 0)(0, y)) = m(0) = 0$$

Thus, M has trivial multiplication. It is now easy to verify that the map

$$B \ltimes M \to X$$

defined by $(b, m) \mapsto e(b) + m$ is an isomorphism of A-algebras over B. The functor

$$\ker(\epsilon) : \mathcal{C}_{ab} \to \mathbf{Mod}_B$$

is then the desired inverse.

$\square$

Remark 3.1.1. It is in general not true that the subcategory of abelian objects in a category is abelian. For example, in **Top**, the abelian objects are the topological abelian groups. But the subcategory is not abelian since continuous bijections need not be homeomorphisms, and thus the category is not balanced.

Corollary 3.1.1. There is an adjunction

$$\mathbf{Mod}_B \underset{B\otimes_A\Omega_{-/A}}{\overset{B\ltimes -}{\underset{\longleftarrow}{\overset{\longrightarrow}{\top}}}} \mathcal{C}$$

Proof. Follows from the standard isomorphism

$$\mathrm{Hom}_B(B \otimes_A \Omega_{X/A}, M) \cong \mathrm{Der}_A(X, M)$$

and Lemma 3.1. □

Effectively, we see that the functor $B\otimes_A\Omega_{-/A}$ is the abelianization functor from $\mathcal{C}$ to $\mathcal{C}_{ab}$, left adjoint to the forgetful functor. The next step is to find the correct homotopical setting to derive the functor. Recall that for the abelian category Mod_R, one computes derived functors via projective resolution in $\mathrm{Ch}(R)$. But now $\mathcal{C} = (\mathrm{Alg}_A)/B$ carries multiplicative structure, and the module of Kähler differentials sees such multiplicative structure. A natural first candidate would be replacing $\mathrm{Ch}(R)$ with the category of commutative differential graded algebras. It turns out that using CDGA works only in characteristic zero. Instead, we simplicial methods work in full generality.

Theorem 3.2 (Model Structure on $s\mathrm{Alg}_R$)**.** The category of simplicial commutative R-algebras $s\mathrm{Alg}_R$ admits a model structure where a morphism $f : A_\bullet \to B_\bullet$ is

1. a weak equivalence is the underlying map of simplicial R-modules is a weak equivalence;
2. a fibration if the underlying map of simplicial R-modules is a fibration

The abelianization functor extends levelwise to a functor between the corresponding simplicial categories

$$B \otimes_A \Omega_{-/A} : s\mathcal{C} \to s\mathbf{Mod}_B$$

Since the right adjoint $B \ltimes -$ preserves weak equivalences and fibrantions, the adjunction in Proposition 3.1.1 promotes to a Quillen adjunction between simplicial model categories.

Definition 3.2.1. The **Cotangent Complex** $L_{B/A}$ is defined to be the total left derived functor of the abelianization functor evaluated at B.

By the general theory of model categories, the cotangent complex can be computed by choosing a cofibrant replacement $P_\bullet \to B$ in $s\mathcal{C}$, and then applying the abelianization functor levelwise to get a simplicial B-module $B \otimes_A \Omega_{P_\bullet/A}$. The cotangent complex is then given by the associated chain complex of B-modules via the Dold-Kan correspondence.

Definition 3.2.2. Suppose M is B-module. The **André-Quillen cohomology** of the A-algebra B with coefficients in M is defined to be

$$D^i(B/A, M) := H^i(\mathrm{Hom}_B(L_{B/A}, M))$$

The **André-Quillen homology** is defined to be

$$D_i(B/A, M) := H_i(L_{B/A} \otimes_B M)$$

3.2 Properties of the Cotangent Complex

Note given ordinary ring homomorphism $R \to S$, there is a Quillen adjunction

$$S \otimes_R - : \mathrm{sAlg}_R \underset{\longleftarrow}{\overset{\longrightarrow}{}} \mathrm{sAlg}_S : \mathrm{res}_R$$

since the right adjoint res_R preserves fibrations and weak equivalences. Therefore, given a commutative diagram of ring homomorphisms

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array}$$

there is a natural morphism in the homotopy category of simplicial B-modules

$$B \otimes_A L_{A/R} \to L_{B/S}$$

This can be realized by first choosing cofibrant replacements $P \to A$ in $s\mathrm{Alg}_R$. Since $R \to P$ is a cofibration, the base change $S \to S \otimes_R P$ is also a cofibration, and $S \otimes_R P$ is a cofibrant S-algebra. We may then factor the induced map $S \otimes_R P \to B$ as a cofibration followed by a trivial fibration

$$S \otimes_R P \xrightarrow{i} Q \xrightarrow{p} B$$

where Q is automatically a cofibrant S-algebra since $S \otimes_R P$ is. We now have a commutative diagram of simplicial rings

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ P & \longrightarrow & Q \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array}$$

which induces the desired morphism of simplicial B-modules

$$B \otimes_A (A \otimes_P \Omega_{P/R}) \to B \otimes_Q \Omega_{Q/S}$$

Theorem 3.3 (Flat Base Change)**.** If A, B are R-algebras such that $\mathrm{Tor}_i^R(A, B) = 0$ for all $i > 0$, then there is a natural isomorphism

$$B \otimes_R L_{A/R} \to L_{A \otimes_R B/B}$$

of $A \otimes_R B$-modules.

Proof. The morphism is induced by the diagram of ring homomorphism

$$\begin{array}{ccc} R & \longrightarrow & A \\ \downarrow & & \downarrow \\ B & \longrightarrow & A \otimes_R B \end{array}$$

Let $P \to A$ be a cofibrant replacement in $s\mathrm{Alg}_R$. The vanishing of the Tor groups implies that the base change

$$P \otimes_R B \to A \otimes_R B$$

is a weak equivalence. We then have a sequence of weak equivalences

$$\begin{aligned} L_{A \otimes_R B/B} &\cong (A \otimes_R B) \otimes_{P \otimes_R B} \Omega_{P \otimes_R B/B} \\ &\cong (A \otimes_R B) \otimes_P \Omega_{P/R} \\ &\cong B \otimes_R (A \otimes_P \Omega_{P/R}) \\ &\cong B \otimes_R L_{A/R} \end{aligned}$$

Corollary 3.3.1. Let A, B be R-algebras, with B flat over R, and M a $A \otimes_R B$-module. Then there is a natural isomorphism

$$D^i(A \otimes_R B/B, M) \cong D^i(A/R, M)$$

□

Theorem 3.4 (Transitivity Sequence)**.** Given a sequence of ring homomorphisms

$$A \xrightarrow{f} B \xrightarrow{g} C$$

there is a natural distinguished triangle in the derived category of C-modules

$$C \otimes_B L_{B/A} \to L_{C/A} \to L_{C/B} \to C \otimes_B L_{B/A}[1]$$

Proof. First choose cofibrant replacement $P \to B$, and factorize the composition $P \to B \to C$ as a free morphism followed by a trivial fibration

$$P \to Q \to C$$

in $s\mathrm{Alg}_A$. Equivalently, Q is the cofibrant replacement of B in $s\mathrm{Alg}_P$. We then get a commutative diagram of simplicial rings

$$\begin{tikzcd}
& P \arrow[r] \arrow[d] & Q \arrow[d] \arrow[ddr, bend left] & \\
A \arrow[ur] \arrow[r] & B \arrow[r] \arrow[drr, bend right] & B \otimes_P Q \arrow[dr] & \\
& & & C
\end{tikzcd}$$

For a C-module N, the top row induces a short exact sequence of derivations

$$0 \to \mathrm{Der}_P(Q, N) \to \mathrm{Der}_A(Q, N) \to \mathrm{Der}_A(P, N) \to 0$$

where exactness on the right follows from the fact that $P \to Q$ is a free morphism, hence admits a section in $s\mathrm{Alg}_A$. The short exact sequence of derivations then induces a short exact sequence of simplicial C-modules

$$0 \longrightarrow C \otimes_B (B \otimes_P \Omega_{P/A}) \longrightarrow C \otimes_Q \Omega_{Q/A} \longrightarrow C \otimes_{Q \otimes_P B} \Omega_{Q \otimes_p B/B} \longrightarrow 0$$

We can immediately identify the left two terms as $C \otimes_B L_{B/A}$ and $L_{C/B}$, so it remains to identify the last term as $L_{C/B}$. The identification follows from showing that $Q \otimes_P B \to C$ is a cofibrant replacement of C in $s\mathrm{Alg}_B$. Since $B \to B \otimes_P Q$ is the base change of the cofibration $P \to Q$, we see that $Q \otimes_P B$ is a cofibrant B-algebra. Moreover, since the model structure of $s\mathrm{Ring}$ is left proper, $Q \to B \otimes_P Q$ is a weak equivalence, and by two-out-of-three, so is $Q \otimes_P B \to C$. Thus, we have the desired identification.

□

Corollary 3.4.1. Given a sequence of ring homomorphisms

$$A \xrightarrow{f} B \xrightarrow{g} C$$

and a C-module M, there is a long exact sequence on cohomology

$$D^0(C/B, M) \to D^0(C/A, M) \to D^0(B/A, M) \to D^1(C/B, M) \to \cdots$$

3.3 Lichtenbaum-Schlessinger Functors

Suppose the base ring is a field of characteristic zero. Then the category of simplicial commutative k-algebras is Quillen equivalent to the category of commutative differential graded k-algebras via the normalization functor. In particular, it is possible to compute the cotangent complex via explicit free resolutions using commutative differential graded algebras. Suppose $K_\bullet$ is such a resolution of B as an A-algebra, the cotangent complex $L_{B/A}$ is given by the complex of B-modules

$$L_{B/A} := \Omega_{K_\bullet/A} \otimes_A B$$

To explicit construct such a resolution, we first recall the Koszul resolution. Suppose B is an A-algebra of finite type. Choose a surjection $f : R = A[y_1, ..., y_j] \to B$, and further choose a generating sequence $(x_1, ..., x_n)$ of $I = \ker(f)$. There is then a standard construction of a free differential graded A-algebra $K_\bullet$, called the Koszul complex, admitting an augmentation map to $B \cong R/I$. Explicitly, the Koszul complex is given by

$$0 \to \wedge^n R^n \to ... \to \wedge^2 R^n \xrightarrow{d_2} R^n \xrightarrow{d_1} R \to 0$$

where d_1 is the R-module homomorphism sending the i-th basis element to x_i, and d_2 is defined by the **Koszul relations**

$$d_2(e_i \wedge e_j) = d_1(e_i)e_j - d_1(e_j)e_i$$

If the defining sequence $(x_1, ..., x_n)$ is a regular, then it is a classic result that the Koszul complex is exact, hence a free CDGA resolution of B as an A-algebra. In general, the Koszul complex need not be exact, but we may fix this by adding more generators in higher degrees according to the cohomology. Clearly, $H^0(K_\bullet) = R/I$. Let $Q = \ker(d_1)$ and $F_0 = \mathrm{im}(d_2)$. We may enlarge the complex by adjoining a free R-module, with basis corresponding to a choice of generators $H^1(K_\bullet) = Q/F$. Extend the differentials accordingly, and we fix the exactness at degree 1. Repeating this process inductively, we eventually get a free CDGA resolution $T_\bullet \to B$.

Definition 3.4.1. The **Lichtenbaum-Schlessinger cotangent complex** of the free CDGA resolution $T_\bullet \to B$ truncated at degree 2. Explicitly, it is given by the complex of B-modules

$$Q/F \otimes_A B \to R^n \otimes_R B \to \Omega_{R/A} \otimes_R B \to 0$$

so we have a short exact sequence

$$0 \to I \to R \to B \to 0$$

Then, choose a free R-module F and a R-module surjection $F \to I$, and let Q be the kernel, so we have a short exact sequence

$$0 \to Q \to F \xrightarrow{j} I \to 0$$

Now define the F_0 as the submodule of F generated by elements of the form $j(a)b - j(b)a$ for all $a, b \in F$. Since j is a R-module homomorphism, we see $j(F_0) = 0$ so that $F_0 \subseteq Q$.

Definition 3.4.2. The **cotangent complex** of B-modules is defined to be the complex

$$L_2 \xrightarrow{d_2} L_1 \xrightarrow{d_1} L_0$$

where $L_2 = Q/F_0$, $L_1 = F \otimes_R B \cong F/IF$ and $L_0 = \Omega_{R/A} \otimes_R B$. The differential d_2 is induced by the inclusion $Q \to F$. The differential d_1 is defined by the composition

$$F/IF \xrightarrow{j} I/I^2 \xrightarrow{d} \Omega_{R/A} \to \Omega_{R/A} \otimes_R B$$

Definition 3.4.3. For any B-module M, define

$$T^i(B/A, M) := H^i(\mathrm{Hom}_B(L_\bullet, M))$$

A Model Categories

A.1 Basic Definitions

We follow the convention in [3] for the definition of model categories. We will assume that all categories are complete and cocomplete.

Definition A.0.1. Suppose $\mathcal{C}$ is a category. A **functorial factorization** (α, β) in $\mathcal{C}$ is the data of two functors $\alpha, \beta : \mathrm{Mor}\ \mathcal{C} \to \mathrm{Mor}\ \mathcal{C}$ such that

$$f = \beta(f) \circ \alpha(f)$$

for all $f \in \mathrm{Mor}\ \mathcal{C}$.

Definition A.0.2. Suppose $i : A \to B$ and $p : X \to Y$ are morphisms in $\mathcal{C}$. We say i has the **left lifting property** with respect to p and p has the **right lifting property** with respect to i if, for every commutative diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"'] & X \arrow[d, "p"] \\ B \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

the dashed lift exists making the entire diagram commute.

Definition A.0.3. A **model structure** on a category $\mathcal{C}$ is three distinguished classes of morphisms

1. weak equivalences;
2. cofibrations;
3. fibrations

and two functorial factorizations (α, β) and (γ, δ) satisfying the following axioms:

- (2-out-of-3) If two of the three morphisms $f, g, g \circ f$ are weak equivalences, so is the third.
- (Retract) Each of the three classes of morphisms is closed under retracts.
- (Lifting) Call a morphism a **trivial (co)fibration** if it is both a (co)fibration and a weak equivalence. Then, trivial cofibrations have the left lifting property with respect to fibrations, and cofibrations have the left lifting property with respect to trivial fibrations.
- (Factorization) For any morphism f, (α, β) factorizes f into a cofibration followed by a trivial fibration, and (γ, δ) factorizes f into a trivial cofibration followed by a fibration.

Proposition A.0.1. Suppose $\mathcal{C}$ is a model category. For any fixed object A in $\mathcal{C}$, the over category $\mathcal{C}/A$ admits a model structure.

Proof. There is a natural forgetful functor $U : \mathcal{C}/A \to \mathcal{C}$ that forgets the structure map to A. We define a morphism in $\mathcal{C}/A$ to be a weak equivalence, cofibration, or fibration if its image under the forgetful functor is such in $\mathcal{C}$. Functorial factorization are also obtained by first factoring in $\mathcal{C}$, then adding the missing structure morphism by composition. The axioms of model categories can then be checked directly from the corresponding axioms in $\mathcal{C}$. $\square$

Definition A.0.4. Suppose $\mathcal{C}$ is category with a distinguished class of morphisms called weak equivalences. Then, the **homotopy category** $\mathrm{Ho}(\mathcal{C})$ is the category obtained by formally inverting all weak equivalences in $\mathcal{C}$.

The standard construction of formally inverting all weak equivalences in $\mathcal{C}$ makes the homotopy category hard to work with directly: the morphsims in $\mathrm{Ho}(\mathcal{C})$ are formal zig-zags, and we do not know a priori if $\mathrm{Ho}(\mathcal{C})$ is locally small, even if $\mathcal{C}$ is. However, model categories admit a convenient description of morphisms in the homotopy category via cofibrant and fibrant replacements.

Definition A.0.5. Suppose $\mathcal{C}$ is a model category, then an object in $\mathcal{C}$ is called **fibrant** if the morphism to the terminal object is a fibration, and **cofibrant** if the morphism from the initial object is a cofibration.

By assumption, our category $\mathcal{C}$ is complete and cocomplete, so it admits an initial object I and a terminal object $*$. Applying functorial factorization (α, β) to the morphism $I \to X$ for any object X, we get factorization

$$\begin{tikzcd} & QX \arrow[dr, "q_X"] & \\ I \arrow[ur] \arrow[rr, "i"'] & & X \end{tikzcd}$$

where QX by definition is a cofibrant, and q_X is a weak equivalence. Similarly, applying (γ, δ) to the morphism $X \to *$ produces

$$\begin{tikzcd} & RX \arrow[dr] & \\ X \arrow[ur, "r_X"] \arrow[rr, "j"'] & & * \end{tikzcd}$$

where RX is fibrant, and r_X is a weak equivalence.

Definition A.0.6. The functor $Q : \mathcal{C} \to \mathcal{C}_c$ is called the **cofibrant replacement functor**, and the functor $R : \mathcal{C} \to \mathcal{C}_f$ is called the **fibrant replacement functor.**

Lemma A.1. Let $\mathcal{C}$ be a model category, and C_c, (resp. $\mathcal{C}_f, \mathcal{C}_{cf}$) be the subcategories of cofibrant(resp. fibrant, cofibrant-fibrant) objects. Then, the inclusion functors induce equivalences of categories

$$\mathrm{Ho}\mathcal{C}_{cf} \to \mathrm{Ho}\mathcal{C}_c \to \mathrm{Ho}\mathcal{C}$$

and

$$\mathrm{Ho}\mathcal{C}_{cf} \to \mathrm{Ho}\mathcal{C}_f \to \mathrm{Ho}\mathcal{C}$$

Proof. We will only show the equivalence $\mathrm{Ho}\mathcal{C}_c \to \mathrm{Ho}\mathcal{C}$, since the other cases are similar. Note that the pair of functors

$$\begin{tikzcd} \mathcal{C}_c \arrow[r, shift left, "i"] & \mathcal{C} \arrow[l, shift left, "Q"] \end{tikzcd}$$

both preserves weak equivalences, so they descend to functors on the homotopy categories. Moreover, the morphism $QX \xrightarrow{q_X} X$ gives us the natural transformations $Q \circ i \to \mathrm{Id}_{\mathcal{C}_c}$ and $i \circ Q \to \mathrm{Id}_{\mathcal{C}}$, and descends to an isomorphism of functors on the homotopy category since it is a weak equivalence. □

Definition A.1.1. Suppose A, B are objects in a model category $\mathcal{C}$. A **cylinder object** for A is a factorization of the fold map

$$\begin{tikzcd} A \coprod A \arrow[d, "\mathrm{fold}"'] \arrow[dr, "i_0+i_1"] & \\ A & A' \arrow[l, "s"'] \end{tikzcd}$$

where $i_0 + i_1$ is a cofibration and s is a weak equivalence. Dually, a **path object** for B is a factorization of the diagonal map

$$\begin{tikzcd} B \times B & \\ B \arrow[u, "\mathrm{diag}"'] \arrow[r, "r"'] & B' \arrow[ul, "(p_0,p_1)"'] \end{tikzcd}$$

where (p_0, p_1) is a fibration and r is a weak equivalence.

Note that applying functorial factorization to the fold and diagonal maps give us canonical choices of cylinder and path objects, denoted by $X \times I$ and X_I. The notation is suggestive of the topological interval $I = [0,1]$, and that the cylinder and path object are analogs of the mapping cylinder and path space constructions in topology. They will provide us with notions of homotopies between morphisms in a model category.

Definition A.1.2. Suppose $f, g : A \to B$ are morphisms in a model category $\mathcal{C}$. A **left homotopy** from f to g is a morphism $H : A' \to B$, where A' is a cylinder object for A, such that $Hi_0 = f$ and $Hi_1 = g$. Dually, a **right homotopy** from f to g is a morphism $K : A \to B'$, where B' is a path object for B, such that $p_0 K = f$ and $p_1 K = g$. We say f, g are **homotopic** if there exists a left and right homotopy between them.

We will write $f \sim g$ if f and g are homotopic.

Proposition A.1.1. Homotopy is an equivalence relation on $\mathrm{Hom}_{\mathcal{C}}(A, B)$ if A is cofibrant and B is fibrant.

Proposition A.1.2. In the subcategory of cofibrant-fibrant objects $\mathcal{C}_{cf}$, weak equivalences and homotopy equivalences coincide.

We summarize the results in the following theorem:

Theorem A.2. Suppose $\mathcal{C}$ is a model category

A.2 Derived Functors

A.3 Simplicial Modules and Rings

Theorem A.3 (Dold-Kan)**.** There is an equivalence of categories

$$s\mathrm{Mod}_R \rightleftarrows \mathrm{Ch}_*(R)$$

The Dold-Kan correspondence, and the known model structure on $\mathrm{Ch}_*(R)$ induces a model structure on $s\mathrm{Mod}_R$. In fact, we can replace Mod_R with any abelian category $\mathcal{A}$, and get an equivalence between $s\mathcal{A}$ and the category of non-negatively graded chain complexes in $\mathcal{A}$.

Now suppose $R = k$ is a field. To incorporate multiplicative structure, we can first try promoting $\mathrm{Ch}(R)$ to the category of commutative differential graded k-algebras, denoted by CDGA_k. There is a free-forgetful adjunction between $\mathrm{Ch}_*(k)$ and CDGA_k:

$$U : \mathrm{CDGA}_k \rightleftarrows \mathrm{Ch}_*(k) : S$$

where U sends a commutative differential graded algebra to its underlying chain complex, and S sends a chain complex to the free commutative differential graded algebra generated by it. The obvious candidate for weak equivalences in both categories are the quasi-isomorphisms. However, the symmetric algebra functor does not preserve weak equivalences when $\mathrm{char}(k) > 0$.

Example A.3.1. Let $k = \mathbb{F}_2$. Note that graded commutativity reduces to strict commutativity in characteristic 2. Consider the acyclic chain complex $D(n)$ given by

$$0 \to k \xrightarrow{\mathrm{id}} k \to 0$$

with non-trivial terms in degree n and $n-1$. The CDGA generated by the trivial acyclic complex is the initial CDGA, which has 1 copy of k concentrated in degree 0. However, the free CDGA $S(D(n))$ has non-trivial homology in degree $2n:$ pick any non-zero element $x \in k$ in degree n, then x^2 is a non-zero element in degree $2n$ with $d(x^2) = 2xd(x) = 0$, so it is a cycle. On the other hand, $d(x^i y^j)$ never contains the homogeneous summand of the form x^n, so x^2 is a non-zero cohomology class.

The example demonstrates the difficulty in setting up a model structure on CDGA_k (with quasi isomorphisms as weak equivalences) in positive characteristic. Instead, it is easier to transfer the model structure from $s\mathrm{Mod}_R$ to the category of commutative simplicial k-algebras $s\mathrm{Alg}_R$.

Let $\mathcal{C}$ be a bicomplete category, together with an adjunction

$$\begin{tikzcd} s\mathcal{C} \arrow[r, shift left, "G"] \arrow[r, phantom, "\top"] & \mathrm{sSet} \arrow[l, shift left, "F"] \end{tikzcd}$$

The following theorem provides sufficient conditions for transferring the standard model structure on sSet to $\mathcal{C}$.

Theorem A.4. Suppose G commutes with filtered colimits. Then, there is a simplicial closed model structure on $s\mathcal{C}$ where a morphism f is

1. a weak equivalence if Gf is a weak equivalence in sSet;
2. a fibration if Gf is a fibration in sSet;
3. a cofibration if it has the left lifting property with respect to all trivial fibrations.

if and only if the following hypothesis holds: every cofibration with the left lifting property with respect to fibrations is a weak equivalence.

The proof goes as follows: since G commutes with filtered colimits, the adjoint F preserves the small objects in sSet with respect to the standard generating (trivial) cofibrations. Thus, we can apply the small object argument to get functorial factorizations in $s\mathcal{C}$. The rest of the axioms can then be checked directly. The details can be found in [1]. In nice situations, the additional hypothesis in the theorem can be verified by the following lemma:

Lemma A.5. Under the same setup as above, suppose $s\mathcal{C}$ admits a fibrant replacement functor, i.e for every object A in $s\mathcal{C}$, there is a weak equivalence $A \to RA$ where RA is fibrant. Then, every cofibration with the left lifting property with respect to fibrations is a weak equivalence.

In our case where $\mathcal{C} = \mathbf{Alg}_R$, the forgetful functor

$$G : s\mathbf{Alg}_R \to \mathrm{sSet}$$

factors through the subcategory of simplicial groups, so every object is automatically fibrant. In particular, we can take the fibrant replacement functor to be the identity functor. Therefore, applying Theorem A.4 gives us the desired model structure on $s\mathbf{Alg}_R$. However, in order to make computations possible, we need an explicit description of the cofibrations.

Definition A.5.1. A morphism of simplicial rings $f : R \to S$ is **free** if there is are subsets $C_q \subset S_q$ such that

1. for every surjective map of sets $\eta : [p] \to [q]$, we have $\eta^*(C_q) \subset C_p$;
2. S_q is the free R_q-algebra generated by C_q.

Proposition A.5.1. A morphism in $s\mathrm{Alg}_R$ is a cofibration if and only if it is a retract of a free morphism.

Proposition A.5.2. A morphism in $s\mathrm{Alg}_R$ can be factored as a free morphism followed by a trivial fibration.

Theorem A.6. Let R be a commutative ring and $s\mathbf{Alg}_R$ be the category of commutative simplicial R-algebras. Then, $s\mathbf{Alg}_R$ admits a model structure where a morphism $f : A_\bullet \to B_\bullet$ is

1. a weak equivalence if $\pi_* X \to \pi_* Y$ is a weak equivalence.
2. a fibration if $X \to \pi_0 X \times_{\pi_0 Y} Y$ is a surjetion.

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